

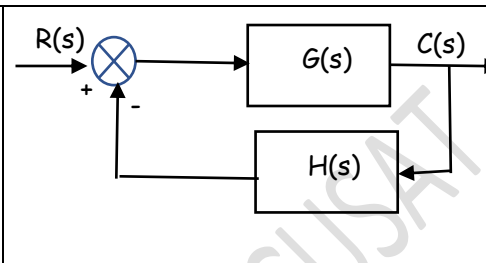
8. ROOT LOCUS METHOD

Root locus is a graphical representation of closed loop poles as a system parameter is varied. It is a powerful method of analysis of stability and transient response. We have learnt to design a first order or second order system for their transient responses by choosing appropriate gain and other system parameters. The real power of root locus technique lies in its ability to provide solutions for higher order systems.

Consider the system shown in figure. The closed loop transfer function

$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

The characteristic equation of this system is $1 + G(s)H(s) = 0$ or $G(s)H(s) = -1 + j0$



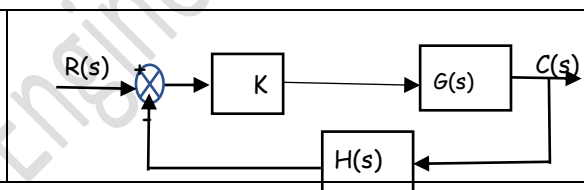
In most of the cases, $G(s)H(s)$ will be a polynomial in the complex variable 's'. Therefore, we can write the magnitude condition as $|G(s)H(s)| = 1$

and the angle condition as $\angle G(s)H(s) = \pm(2k+1)\pi$ (odd multiples of 180°) $k=0,1,2,\dots$

The values of s that satisfies both the angle and magnitude conditions are the roots of the characteristic equation or the closed-loop poles.

In most cases, the forward path involves a gain parameter K so that the transfer function is

$$T(s) = \frac{C(s)}{R(s)} = \frac{KG(s)}{1 + KG(s)H(s)}$$



And the characteristic equation becomes $1 + KG(s)H(s) = 0$

The locus of the roots of this characteristic equation when the parameter K is varied from zero to infinity is known as the root locus.

Root Locus Procedure

Step 1: Write the characteristic equation in the form $1 + K \frac{\prod_{i=1}^m (s+z_i)}{\prod_{j=1}^n (s+p_j)} = 0$

For example consider $1 + K \frac{(s+1)}{s(s+2)(s+4)^2} = 0$

Step 2: Locate the poles of the closed loop transfer function

Root locus is the locus of the roots of the characteristic equation (poles) when the parameter K varies from 0 to ∞

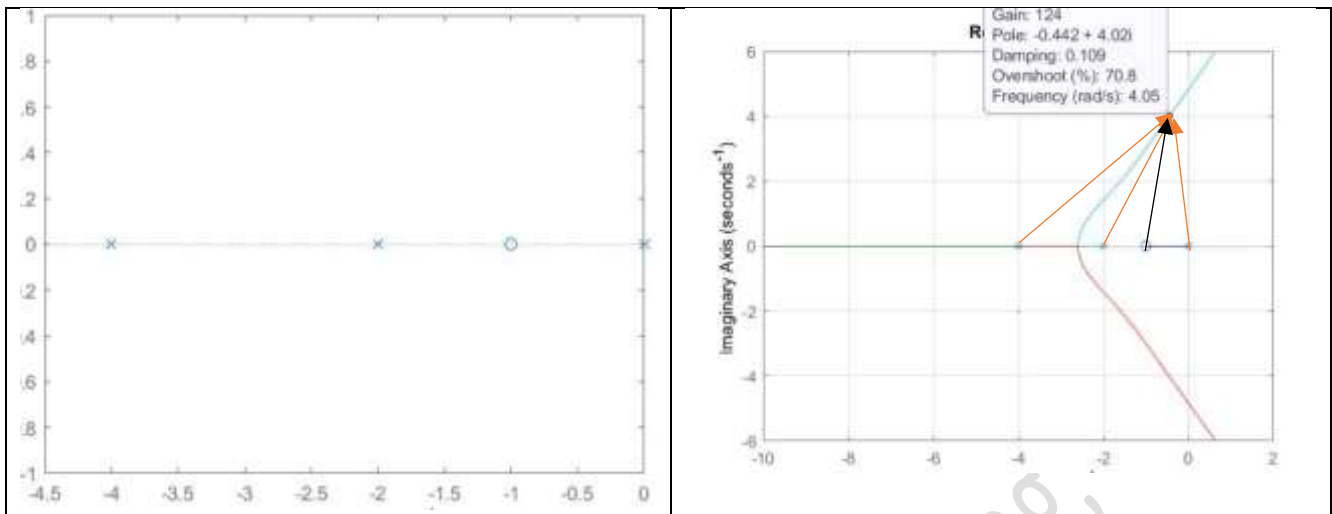
When $K = 0$, $-\frac{\prod_{j=1}^n (s+p_j)}{\prod_{i=1}^m (s+z_i)} = 0 \gg \prod_{j=1}^n (s+p_j) = 0$, the roots are simply the poles of $G(s)H(s)$.

When $K = \infty$, $-\frac{\prod_{j=1}^n (s+p_j)}{\prod_{i=1}^m (s+z_i)} = \infty \gg \prod_{i=1}^m (s+z_i) = 0$, the roots are simply the zeros of $G(s)H(s)$.

Therefore, the root locus starts from the poles and end at the zeros of $G(s)H(s)$ as K varies from 0 to infinity. If the number of poles are n and number of zeros are m, usually n will be greater than m. Then $n - m$ zeros will lie at infinity.

The poles are $p_j = 0, -2, -4, -4$ and zeros are $z_i = -1$. Here we have four poles ($n = 4$) and one zero ($m = 1$). The root locus will start from three poles and end at the zeros. Three branches of the root loci will go to the zeros at infinity ($n - m = 3$)

Step 3: Locate the segments of the real axis that are part of the root loci. We know that any point on the root locus will satisfy the angle condition. The poles and root locus of the above example is shown in the figure below.



$$G(s)H(s) = \frac{(s+1)}{s(s+2)(s+4)^2}$$

$$\angle G(s)H(s) = \angle \frac{(s+1)}{s(s+2)(s+4)^2} = \angle (s+1) - \angle s - \angle (s+2) - \angle (s+4) - \angle (s+4)$$

Let us see if the point $s = -0.44 + j4.02$ is a point on the root locus. Substitute this value of s in the above expression.

$$\angle (s+1) = \angle (-0.44 + j4.02) + 1 = \angle 0.56 + j4.02$$

$$= \tan^{-1} \frac{4.02}{0.56} = 82.07^\circ$$

$$\angle (s) = \angle (-0.44 + j4.02) = \angle -0.44 + j4.02$$

$$= \tan^{-1} \frac{4.02}{-0.44} = -83.75^\circ \text{ (or } 96.25^\circ)$$

$$\angle (s+2) = \angle (-0.44 + j4.02) + 2 = \angle 1.56 + j4.02$$

$$= \tan^{-1} \frac{4.02}{1.56} = 68.79^\circ$$

$$\angle (s+4) = \angle (-0.44 + j4.02) + 4 = \angle 3.56 + j4.02$$

$$= \tan^{-1} \frac{4.02}{3.56} = 48.47^\circ$$

$$\angle G(s)H(s) = 82.07 - 96.25 - 68.79 - 48.47 - 48.47 = -180$$

Which is an odd multiple of 180. Hence the point $(-0.44 + j4.02)$ is on the root locus.

Now let us consider a point on the real axis (say $s = -0.5$)

$$\angle (s+1) = \angle (-0.5 + j0) + 1 = \angle 0.5 + j0 = \tan^{-1} \frac{0}{0.5} = 0^\circ$$

$$\angle (s) = \angle (-0.5 + j0) = \tan^{-1} \frac{0}{-0.5} = 180^\circ$$

$$\angle (s+2) = \angle (-0.5 + j0) + 2 = \angle 1.5 + j0 = \tan^{-1} \frac{0}{1.5} = 0^\circ$$

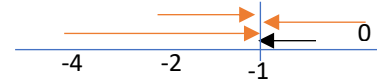
$$\angle (s+4) = \angle (-0.5 + j0) + 4 = \angle 3.5 + j0 = \tan^{-1} \frac{0}{3.5} = 0^\circ$$

$$\angle G(s)H(s) = 0 - 180 - 0 - 0 - 0 = -180$$

Which is an odd multiple of 180. Hence the point $(s = -0.5)$ is on the root locus. By this argument, any point between pole at 0 and zero at -1 will satisfy the angle criterion. Hence the real axis segment between $s=0$ and $s=-1$ will form part of the root loci.

Now let us consider another point on the real axis (say $s = -1.5$)

$$\begin{aligned}\angle(s+1) &= \angle(-1.5 + j0) + 1 = \angle -0.5 + j0 = \tan^{-1} \frac{0}{-0.5} = 180^\circ \\ \angle(s) &= \angle(-1.5 + j0) = \tan^{-1} \frac{0}{-0.5} = 180^\circ \\ \angle(s+2) &= \angle(-1.5 + j0) + 2 = \angle 0.5 + j0 = \tan^{-1} \frac{0}{0.5} = 0^\circ \\ \angle(s+4) &= \angle(-1.5 + j0) + 4 = \angle 2.5 + j0 = \tan^{-1} \frac{0}{2.5} = 0^\circ\end{aligned}$$

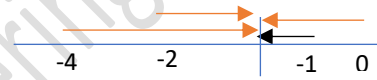


$$\angle G(s)H(s) = 180 - 180 - 0 - 0 - 0 = 0$$

Which is not an odd multiple of 180. Hence the point ($s = -1.5$) is not on the root locus. By this argument, any point between zero at -1 and pole at -2 will not satisfy the angle criterion. Hence the real axis segment between $s = -2$ and $s = -1$ will not form part of the root loci.

Consider another point on the real axis (say $s = -2.5$)

$$\begin{aligned}\angle(s+1) &= \angle(-2.5 + j0) + 1 = \angle -1.5 + j0 = \tan^{-1} \frac{0}{-1.5} = 180^\circ \\ \angle(s) &= \angle(-2.5 + j0) = \tan^{-1} \frac{0}{-2.5} = 180^\circ \\ \angle(s+2) &= \angle(-2.5 + j0) + 2 = \angle -0.5 + j0 = \tan^{-1} \frac{0}{-0.5} = 180^\circ \\ \angle(s+4) &= \angle(-2.5 + j0) + 4 = \angle 1.5 + j0 = \tan^{-1} \frac{0}{1.5} = 0^\circ\end{aligned}$$

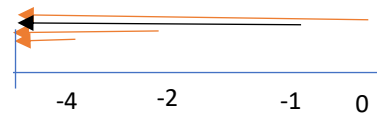


$$\angle G(s)H(s) = 180 - 180 - 180 - 0 - 0 = -180$$

Which is an odd multiple of 180. Hence the point ($s = -2.5$) is on the root locus. By this argument, any point between pole at -2 and poles at -4 will satisfy the angle criterion. Hence the real axis segment between $s = -4$ and $s = -2$ will form part of the root loci.

Consider another point on the real axis (say $s = -4.5$)

$$\begin{aligned}\angle(s+1) &= \angle(-4.5 + j0) + 1 = \angle -3.5 + j0 = \tan^{-1} \frac{0}{-3.5} = 180^\circ \\ \angle(s) &= \angle(-4.5 + j0) = \tan^{-1} \frac{0}{-4.5} = 180^\circ \\ \angle(s+2) &= \angle(-4.5 + j0) + 2 = \angle -2.5 + j0 = \tan^{-1} \frac{0}{-2.5} = 180^\circ \\ \angle(s+4) &= \angle(-4.5 + j0) + 4 = \angle -0.5 + j0 = \tan^{-1} \frac{0}{-0.5} = 180^\circ\end{aligned}$$



$$\angle G(s)H(s) = 180 - 180 - 180 - 180 - 180 = 3 \times (-180)$$

Which is an odd multiple of 180. Hence the point ($s = -4.5$) is on the root locus. By this argument, any point between pole at -4 and $s = -\infty$ will satisfy the angle criterion. Hence the real axis segment between $s = -4$ and $s = -\infty$ will form part of the root loci.

The real axis segment to the left of odd number of poles and/or zeros will be part of the root locus.

Step 4: Determine the separate loci (SL). Because the loci begin at the poles and end at the zeros, the number of separate loci is equal to the number of poles.

Step 5: Root loci must be symmetrical with respect to the real axis because the complex roots occur as conjugate pairs.

Step 6: The root loci proceed to the zeros at infinity along the asymptotes centered at σ_A and with angles ϕ_A . When the number of zeros m is less than the number of poles n , then $n - m$ sections of loci must end at infinity.

$$\sigma_A = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m}$$

$$\phi_A = \frac{(2k + 1)180^\circ}{n - m}; \quad k = 0, 1, 2, \dots, (n - m - 1)$$

The centroid of the asymptotes can be found out from the characteristic equation $1 + KG(s)H(s) = 0$. $s = z_1 (s - z_1)(s - z_2)$

$$1 + K \frac{\prod_{i=1}^m (s + z_i)}{\prod_{j=1}^n (s + p_j)} = 0$$

$$1 + K \frac{[s^m - (z_1 + z_2 + \dots + z_m)s^{m-1} + \dots + (z_1 z_2 \dots z_m)]}{[s^n - (p_1 + p_2 + \dots + p_n)s^{n-1} + \dots + (p_1 p_2 \dots p_n)]} = 0$$

As $s \rightarrow \infty$; $1 + K \frac{[s^m - (\sum \text{zeros})s^{m-1}]}{[s^n - (\sum \text{Poles})s^{n-1}]} = 0$

$$1 + K \frac{s^{-m}[s^m - (\sum \text{zeros})s^{m-1}]}{s^{-m}[s^n - (\sum \text{Poles})s^{n-1}]} = 0$$

$$1 + K \frac{[1 - s^{-1}(\sum \text{zeros})]}{[s^{n-m} - s^{n-m-1}(\sum \text{Poles})]} = 0$$

The second term in the numerator is very small; we can use the binomial expansion (for $x < 1$)

$$1 - x = (1 + x)^{-1} = \frac{1}{1 + x}$$

$$1 + K \frac{1}{[s^{n-m} - s^{n-m-1}(\sum \text{Poles})][1 + s^{-1}(\sum \text{zeros})]} = 0$$

$$1 + K \frac{1}{[s^{n-m} + s^{n-m-1}(\sum \text{zeros}) - s^{n-m-1}(\sum \text{Poles}) - s^{n-m-2}(\sum \text{Poles})(\sum \text{zeros})]} = 0$$

$$1 + K \frac{1}{[s^{n-m} + s^{n-m-1}(\sum \text{zeros}) - (\sum \text{Poles})]} = 0$$

Again for large value of s , the characteristic equation $1 + K \frac{\prod_{i=1}^m (s + z_i)}{\prod_{j=1}^n (s + p_j)} = 0$ can be approximated as

$$1 + K \frac{(s - \sigma_A)^m}{(s - \sigma_A)^n} = 0 \gg 1 + K \frac{1}{(s - \sigma_A)^{n-m}} = 0 \gg 1 + K \frac{1}{s^{n-m} - \sigma_A(n-m)s^{n-m-1}} = 0$$

Comparing the coefficients in the two equations, we can write

$$-\sigma_A(n - m) = \sum \text{zeros} - \sum \text{Poles}$$

$$\sigma_A = \frac{\sum \text{Poles} - \sum \text{Zeros}}{n - m}$$

If a test point s is selected very far from the origin, $\angle G(s)H(s) = \pm(2k + 1)\pi$

$$\angle K \frac{(s - \sigma_A)^m}{(s - \sigma_A)^n} = \pm(2k + 1)\pi \quad \angle K \frac{1}{(s - \sigma_A)^{n-m}} = \pm(2k + 1)\pi \quad -(n - m) \angle (s - \sigma_A) = \pm(2k + 1)\pi$$

$$\angle (s - \sigma_A) = \pm \frac{(2k + 1)\pi}{n - m}$$

$$\sigma_A = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m} = \frac{(-4 - 4 - 2 - 0) - (-1)}{4 - 1} = \frac{(-10) + 1}{3} = -3$$

$$\phi_A = \frac{(2k + 1)180^\circ}{n - m}; \quad k = 0, 1, 2$$

$$\phi_A = \frac{(2 \times 0 + 1)180^\circ}{3}, \frac{(2 \times 1 + 1)180^\circ}{3}, \frac{(2 \times 2 + 1)180^\circ}{3} = 60^\circ, 180^\circ, 300^\circ$$

Step 7: Determine the Break Away (Break in) Point. At the break away or break in points there will be multiplicity of roots. The value of s at the break away or break in point can be obtained as follows. Write the characteristic equation in the form $K = P(s)$. The roots of the polynomial $\frac{dK}{ds} = 0$ will give the break away or break in point.

The characteristic equation $1 + KG(s)H(s) = 0$ can be written as $1 + K \frac{Y(s)}{X(s)} = 0$

$$f(s) = X(s) + KY(s) = 0$$

This will have multiple roots at points where $\frac{df(s)}{ds} = 0$

$$X'(s) + KY'(s) = 0$$

$$K = -\frac{X'(s)}{Y'(s)}$$

Substituting this equation in $f(s)$, $X(s)Y'(s) - Y(s)X'(s) = 0$

When we solve this equation for s , we will get the location where multiple roots exists.

If we write $K \frac{Y(s)}{X(s)} = -1$ or $K = -\frac{X(s)}{Y(s)}$. If we find $\frac{dK}{ds}$ an equate to zero

$$\frac{dK}{ds} = -\frac{Y(s)X'(s) - Y'(s)X(s)}{Y^2(s)} = 0$$

We get the same equation $Y'(s)X(s) - Y(s)X'(s) = 0$

$$K = -\frac{s(s+2)(s+4)^2}{s+1} = -\frac{s^4 + 10s^3 + 32s^2 + 32s}{s+1}$$

$$\frac{dK}{ds} = -\frac{(s+1)(4s^3 + 30s^2 + 64s + 32) - (s^4 + 10s^3 + 32s^2 + 32s)}{(s+1)^2} = 0$$

$$3s^4 + 24s^3 + 62s^2 + 64s + 32 = 0$$

The roots are $-4, -2.5994, -0.7003 \pm j0.7317$

The break away point is -2.5994

Step 8: Determine the points where the root loci crosses the imaginary axis. This can be found out by use of the Routh's stability criterion

The characteristic equation of the given example is $1 + K \frac{(s+1)}{s(s+2)(s+4)^2} = 0$

$$s(s+2)(s+4)^2 + K(s+1) = 0$$

$$s^4 + 10s^3 + 32s^2 + (32+K)s + K = 0$$

s^4	1	32	K	
s^3	10	32+K		
s^2	$\frac{288-K}{10}$	K		
s^1	$\frac{K^2 - 156K - 9218}{28.8 - 0.1K}$			
s^0	K			

$$\frac{10 \times 32 - 1 \times (32 + K)}{10}$$

$$\frac{(28.8 - 0.1K) \times (32 + K) - 10K}{28.8 - 0.1K}$$

For $K=201.7$, we will have a row of zeros. Then the auxiliary polynomial will be $8.63s^2 + 201.7 = 0$

$$s = \pm j4.83$$

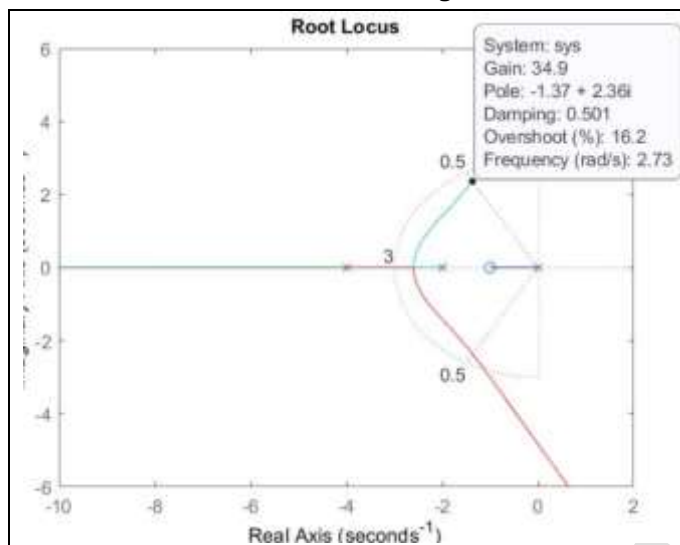
The root loci crosses the imaginary axis at $s = \pm j4.83$

Step 9: Determine the angle of departure from a complex pole and the angle of arrival of the locus at a complex zero, using the angle criterion.

Step 10: Choose test points in such a way that they satisfy the angle criterion. Draw the loci passing through these points.

Step 11: Determine the pair of dominant complex conjugate poles such that the damping ratio is 0.5.

Closed loop poles with $\zeta = 0.5$ lie on a line passing through the origin and making angles $\pm \cos^{-1} \zeta = \pm \cos^{-1}(0.5) = \pm 60^\circ$ with the negative real axis.



The closed loop poles are $-1.37 \pm j2.36$
The value of K that will yield these poles can be obtained from the magnitude condition

$$|KG(s)H(s)| = 1$$

$$K \left| \frac{(s+1)}{s(s+2)(s+4)^2} \right|_{s=-1.37+j2.36} = 1$$

$$K = \left| \frac{s(s+2)(s+4)^2}{s+1} \right|_{s=-1.37+j2.36}$$

$$K = \frac{|s||s+2||s+4||s+4|}{|s+1|} \quad s=-1.37+j2.36$$

$$K = \frac{|-1.37 + j2.36| |-1.37 + j2.36 + 2| |-1.37 + j2.36 + 4| |-1.37 + j2.36 + 4|}{|-1.37 + j2.36 + 1|}$$

$$K = \frac{\sqrt{(-1.37)^2 + (2.36)^2} \cdot \sqrt{(0.63)^2 + (2.36)^2} \cdot \sqrt{(2.63)^2 + (2.36)^2} \cdot \sqrt{(2.63)^2 + (2.36)^2}}{\sqrt{(-0.37)^2 + (2.36)^2}}$$

$$K = \frac{2.724 \times 2.442 \times 3.534 \times 3.534}{2.389} = 34.8$$

With this value of K, the characteristic equation becomes

$$s^4 + 10s^3 + 32s^2 + (32 + K)s + K = 0$$

$$s^4 + 10s^3 + 32s^2 + 66.8s + 34.8 = 0$$

The roots of this equation are $-6.5466, -0.7151, -1.37 + j2.36, -1.37 + j2.36$

References

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3. Control Systems Engineering: Norman S. Nise, 6/e, John Wiley & Sons